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Shear Evolution and Slippage of the Liquid−Liquid Interface over a Liquid-Infused Surface: A Many-Body Dissipative Particle Dynamics Study

Liuzhen Ren, Jiangzhuo Ren, Luyao Bao, Jun Wen, Min Ye, and Haibao Hu*

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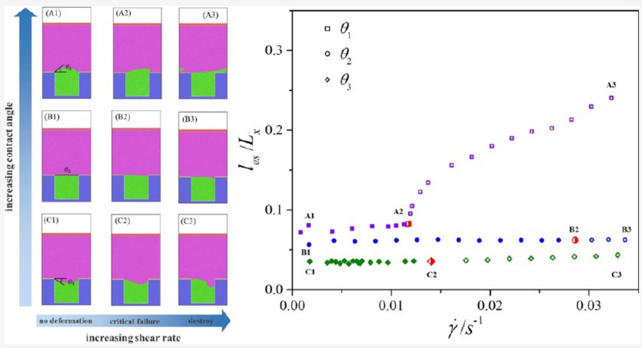
ABSTRACT: The liquid−liquid interface (LLI), which is the key to cause flow slippage and thus promote drag reduction of liquid-infused surfaces (LISs), does suffer from the action of flow shear. In the current study, the transverse many-body dissipative dynamics simulation method is applied to explore the shear evolution of LLI and the corresponding slippage over a periodically grooved LIS. Results show that a relatively small shear rate only induces a slight deformation of LLI and the corresponding effective slippage is dependent on the shear rate. With a further increase of the shear rate, LLI deforms apparently and then the downstream three phase contact line depins to move once the balance between the capillary force and the shear force is broken, which results in an apparent increase of the slippage, specifically for a convex LLI. Compared with a convex LLI or a concave LLI, a flat LLI remains relatively stable under the same shear action, and an increase of the viscosity ratio and a decrease of the LLI fraction can both strengthen the shear resistance of an LLI, while they are less effective to promote the slippage. Consequently, the current results not only indicate that the slippage is related to the interface deflection and the shear rate but also suggest that both the shear resistance and the slippage of LLI should be considered when designing an effective LIS.



■ INTRODUCTION

Underwater drag reduction is significant to reduce energy consumption and enhance traveling speed or distance of underwater vehicles, as well as to minimize the drag losses of fluid transport in pipelines.[1,2] Inspired by the Nepenthes pitcher plants, liquid-infused surfaces (LISs) have been newly proposed to reduce the flow drag since 2011.[3−5] Similar to superhydrophobic surfaces (SHSs),[6−8] LISs are also manufactured with micro- and nanostructures and the low surface energy, but an immiscible liquid lubricant, not the air used for SHSs, is infused within these structures. Since SHSs lack stability as they suffer from the action of the pressure fluctuation and the air dissolution,[9−11] the immiscible lubricant can avoid the deleterious effects that arise with SHSs, which makes LISs relatively stable and thus promising for underwater drag reduction.[12−14] Besides, such LISs are also effective for anticorrosion, anti-icing, condensation, electricity generation, and so on.[15−19]

Until now, the drag-reducing ability of LISs has been validated by torque systems or pressure drop measurements. For example, by applying the plate rheometer to generate a viscous laminar flow in 2014, Solomon et al. first elucidated that the drag reduction of LISs was positively related to the dynamic viscosity ratio between the working fluid and the lubricant fluid, while it increased to a value of 16% as the ratio increased to a value of 260.[20] Later in 2016, Rosenberg et al. achieved a drag reduction of 14% for the best LIS based on a Taylor−Couette apparatus, and first suggested that LISs may enable robust drag reduction in a turbulent flow.[21] Recently, by patterning alternating superhydrophobic strips and superhydrophilic strips over the inner cylinder of a Taylor−Couette apparatus, we have confined stable lubricant rings over the cylindrical surface and thus obtained consequent drag reduction.[22] Meanwhile, by measuring the pressure drop over a lubricant-infused copper capillary, the drag-reducing ability of LIS in such a microchannel was evaluated as well.[23] On the other hand, by directly measuring the frictional drag of a liquid-infused plate in a water tunnel, the drag reduction rate of LIS was directly calculated by Kim et al. in 2023.[24]

When the lubricant is infused into structured substrates, the original solid−liquid interface (SLI) becomes partially replaced

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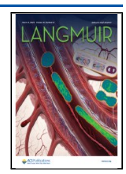

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by a liquid–liquid interface (LLI) on submerged LISs. This LLI formation is crucial for flow slippage and achieving drag reduction effects. For example, by directly measuring the flow profile and then calculating the slip length over a biomimetic LIS, Lee et al. obtained a slip length with a value of 68.02 μm (which corresponds to a drag reduction rate with a value of 18%), while the bare surface only induced a slip length with a value of 0.32 μm and it could not reduce the flow drag.[25] Besides, Chen et al. compared the effective slip length according to the average velocity profile and the value evaluated based on the measured flow rate, and they found that the two methods gave similar results and the slip length was independent of the driving pressure but depended on the viscosity ratio between the working fluid and the lubricant fluid.[26] Besides, Asmolov et al. analytically solved the effective slip length over a flat LLI and demonstrated that it increased with the LLI fraction and the viscosity ratio, whereas the influence of the protrusion angle is rather little.[27] By applying the Stokes flow assumption, Chang et al. suggested a predictive model for slip length and drag reduction rate with a heptane-lubricated surface.[28] Recently, we numerically identified two local slip modes at LLI over an LIS, that was, with an increase of the viscosity ratio, a finite local slip boundary condition would gradually transform into a hybrid slip mode.[29]

Although LISs can avoid the deleterious effects of pressure fluctuations and gas dissolution that arise with SHSs, the action of the flow shear could disable LLI and then weaken the slippage, as exposed for SHSs, which thus makes LISs less effective for drag reduction. For example, Scarratt et al. examined that the slip length over silicone oil film-coated Teflon substrates increased with increasing silicone oil film thickness.[30] Kim et al. found that the lubricant gradually depleted with increasing shear time, and the pressure drop over a liquid-infused channel was increased, i.e., the drag-reducing rate was decreased.[31] By presenting a two-phase Couette flow computational dynamic simulation over LISs, Vega-Sánchez et al. pointed out that the slip length was related to the lubricant retention, and even a slight lubricant loss could decrease the slip to the point of making the lubricant superfluous.[32] Besides, by calculating the slip length according to the pressure drop over a liquid-infused capillary, Yan et al. found that the slip length decreased with increasing shear time, and the corresponding drag reduction effect decreased as well.[33]

Since LLI is significant to maintain the advantageous properties of LISs, it is essential to investigate the shear evolution of LLI and thus to provide the strategy to improve the stability of an LIS. For example, by examining the shear driven drainage of the lubricant over a streamwise rectangular grooved LIS, Wexler et al. demonstrated that a finite length of the surface remained wetted indefinitely and then established the geometric surface parameters governing fluid retention.[34] Jacobi et al. found that the geometry and chemical Marangoni stresses could generate an overflow cascade of the lubricant under such cases and developed general guidelines to estimate the onset of the different stages of the cascade with the aim of providing additional robustness criteria for the design of future LISs.[35] Then, the effects of the viscosity ratio, the streamwise cross-sectional variation, and the surface robustness on the lubricant-retention length were further explored, and a fluid-retention prediction was derived to guide the design of an LIS.[36−38] On the other hand, by studying the meniscus deformation in an outer shear flow-oriented transverse to lubricant-infused rectangular grooves with different depths, Asmolov et al. identified two separate failure scenarios of LLI, which could help avoid this generally detrimental effect in various applications.[39] Lately, the shear-flow-induced lubricant depletion in a dovetail-shaped cavity microchannel under the action of a transverse flow has been numerically examined by Ahuja et al., and three stable and two failure meniscus shapes have been distinguished to help in the design of a robust LIS system.[40]

Although previous studies have investigated the drag-reducing abilities, slip characteristics, and the evolutionary behavior of LLI over LISs, the relationship between the dynamic evolution of LLI and its corresponding slippage remains poorly understood. Furthermore, the effects of LLI deflection on these phenomena require further elucidation. Taking into account the deflection of LLI, whether the meniscus is protruded inside or outside the structures, the evolution of LLIs and the corresponding slip properties of LISs exposed to a shear transverse flow are investigated in the current study. Then, the effects of the deflection of LLI, the viscosity ratio between the working liquid and the lubricant, and the LLI fraction on the deformation and destruction of LIS are all explored, and the corresponding slip characteristics are analyzed. Recently, we have proposed a transverse many-body dissipative particle dynamics (T-MDPD) to effectively simulate the flow property over LISs,[29] and this T-MDPD scheme is thus applied to simulate the shear evolution of LLI in the current study. Note that dissipative particle dynamics (DPD) is a particle-based discrete method, and it can simplify the handling of moving interfaces and capture fine details of fluid behavior in the near-interface region. In a DPD system, particles can represent different phases or components and DPD is thus easy to model interactions between different fluid phases, solid particles, and other complex multibody interactions.[41−43] Specifically, among the extensions of DPD methods, the MDPD scheme is effective to simulate multiphase flows.[41,44,45]

The remainder of this paper is organized as follows. First, the simulation method is described, which consists of the MDPD scheme, simulation setup, and parametrization. Then, by adjusting the deflection of LLI, the deformation, destruction, and corresponding slip characteristics of LIS are investigated and analyzed for various viscosity ratios and LLI fractions. Finally, a summary of this study and our conclusions are presented.

SIMULATION METHODS

T-MDPD Scheme. Different from the standard MDPD (S-MDPD) scheme, where particles interact via three pairwise forces along the interparticle axis and the S-MDPD thermostat cannot effectively control the liquid viscosity,[46−48] a transverse MDPD (T-MDPD) scheme was proposed to effectively tune the viscosity of an MDPD fluid over a large range[49] and applied for simulating the flow over an LIS.[29] Specifically, by adding a lateral friction coefficient to the S-MDPD form, the three pairwise forces, namely, the conservative, dissipative, and random forces in this T-MDPD scheme, are redefined, respectively, as

$$\boldsymbol{F}_{ij}^{\mathrm{C}} = (A_{ij}\omega_{\mathrm{c}}(r_{ij}) + B_{ij}(\bar{\rho}_i + \bar{\rho}_j)\omega_{\mathrm{d}}(r_{ij}))\boldsymbol{e}_{ij} \tag{1}$$

$$\boldsymbol{F}_{ij}^{\mathrm{D}} = -\gamma^{\parallel}\omega_{\mathrm{D}}^{\parallel}(r_{ij})(\boldsymbol{e}_{ij}\cdot\boldsymbol{v}_{ij})\boldsymbol{e}_{ij} - \gamma^{\perp}\omega_{\mathrm{D}}^{\perp}(r_{ij})[\boldsymbol{v}_{ij} - (\boldsymbol{e}_{ij}\cdot\boldsymbol{v}_{ij})\boldsymbol{e}_{ij}] \tag{2}$$

$$\boldsymbol{F}_{ij}^{\mathrm{R}} = \sigma^{\parallel}\omega_{\mathrm{R}}^{\parallel}(r_{ij})(\boldsymbol{e}_{ij}\cdot\boldsymbol{\theta}_{ij})\boldsymbol{e}_{ij} + \sigma^{\perp}\omega_{\mathrm{R}}^{\perp}(r_{ij})[\boldsymbol{\theta}_{ij} - (\boldsymbol{e}_{ij}\cdot\boldsymbol{\theta}_{ij})\boldsymbol{e}_{ij}] \tag{3}$$

Then, the governing equation of the ith particle in the T-MDPD system is given by

$$m_i\frac{\mathrm{d}^2\boldsymbol{r}_i}{\mathrm{d}t^2} = \sum_{i\neq j}\boldsymbol{F}_{ij}^{\mathrm{C}} + \boldsymbol{F}_{ij}^{\mathrm{D}} + \boldsymbol{F}_{ij}^{\mathrm{R}} \tag{4}$$

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In eq 1, for the calculation of F_{ij}^{C}, A_{ij} is the strength of the attractive force and B_{ij} is the strength of the repulsive force. r_{ij} is the distance between particles i and j, and $\boldsymbol{e}_{ij} = (\boldsymbol{r}_i - \boldsymbol{r}_j)/r_{ij}$ is the unit vector in the direction from particle j to i, as r_i represents the position of particle i and r_j represents the position of particle j, respectively. $\omega_c(r)$ $(=(1 - r/r_c))$ and $\omega_d(r)$ $(=(1 - r/r_d))$ are weight functions that vanish for $r > r_c$ and $r > r_d$, respectively. $\overline{\rho}_i = \sum_{i\neq j} \omega_\rho(r_{ij})$ is the local density for each particle within an effective range of r_d, where $\omega_\rho(r) = 105/(16\pi r_d^3)(1 + 3r/r_d)(1 - r/r_d)^3$ is the weight function.

For the calculation of F_{ij}^{D} in eq 2 and F_{ij}^{R} in eq 3, γ and σ are the friction coefficient and the random coefficient, respectively. Specifically, $\gamma^{\parallel}$ and $\sigma^{\parallel}$ are along the direction of e_{ij}, as $\gamma^{\perp}$ and $\sigma^{\perp}$ are perpendicular to this direction. Meanwhile, they satisfy the relation $\sigma^2 = 2\gamma k_B T$ at both these two directions, where k_B is the Boltzmann constant and T is the temperature of the system. With an interaction range of r_c, the weight functions are $\omega_D^{\parallel}(r) = \omega_D^{\perp}(r) = (1 - r/r_c)^2$. The weight functions $\omega_R^{\parallel}(r) = \omega_R^{\perp}(r)$ are chosen such that $\omega_D(r) = [\omega_R(r)]^2$. Besides, $v_{ij} = v_i - v_j$ is the velocity difference, as v_i and v_j are the velocities of particles i and j.

Specifically in eq 3, $\boldsymbol{\theta}_{ij}$ is a three-dimensional time-variant vector whose elements are independent Gaussian random numbers with zero mean and unit variance. This vector satisfies the following conditions:

$$\langle \boldsymbol{\theta}_{ij} \rangle = 0 \tag{5}$$

$$\boldsymbol{\theta}_{ij}(t) \otimes \boldsymbol{\theta}_{kl}(t') = \boldsymbol{I}(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk})\delta(t - t') \tag{6}$$

where $\boldsymbol{I}$ is a 3 × 3 matrix and ⊗ represents the dyad product.

Simulation Setup. The current shear flow over a periodically grooved liquid-infused surface (LIS) is schematically shown in Figure 1.

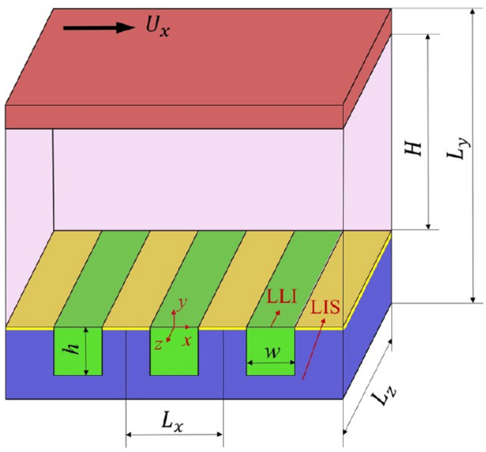

Figure 1. Schematic of the shear flow over a periodically grooved liquid-infused surface (LIS). h and w are the depth and width of the groove, respectively. The red region and the purple region represent the upper wall and the lower wall, respectively. The yellow region over the lower wall represents the topmost layer of the lower wall. The green region and the pink region are filled with lubricant fluid and working fluid, respectively.

The simulation dominates, with the dimensions of L_x (=64.0 in the DPD units) in the x-direction, L_y (which is adjustable as illustrated below) in the y-direction, and L_z (=8.0 in the DPD units) in the z-direction, and a channel is confined between the upper wall and the lower wall. Specifically, a rectangular groove, with a height of h (=32.0 in the DPD units) in the y-direction and width of w (which is adjustable as illustrated below) in the x-direction, is carved inside the lower wall. As a Newtonian liquid, i.e., the working fluid, is confined in the channel, the groove is filled with another immiscible liquid, i.e., the lubricant liquid, to simulate an LIS. By adjusting the position of the upper wall, i.e., the channel height H, the deflection of the liquid−liquid interface (LLI) is controlled as described in our previous study.[29] Note that the location of LLI is defined at the position where the local particle density is equal to half of the sum of the liquid densities in the channel and in the groove. Besides, the viscosity of the lubricant is varied by adjusting the friction coefficient of the lubricant to provide different viscosity ratios (i.e., the ratio of the dynamic viscosity of the working fluid to that of the lubricant fluid) of LLI, as well as the groove width w is also varied to provide different fractions of LLI.

In the simulations, the upper wall, lower wall, and the two kinds of liquids are all constructed from face-centered cubic crystals. With a particle density number of 12.0 in the DPD units, the upper wall consists of 19,980 solid particles. When w is equal to 15.60, 23.57, 31.55, 39.52, and 47.50 in the DPD units, the lower wall, with the same density as the upper wall, consists of 177,876, 152,484, 127,092, 101,700, and 67,428 particles, respectively. Besides, the number of liquid particles in the groove, with a particle density number of 3.24 in the DPD units, is 177,876, 152,484, 127,092, 101,700, and 67,428, respectively, while the working fluid, with a particle density number of 6.74 in the DPD units, always consists of 203,889 particles in the channel. Specifically, a no-slip boundary condition is applied at the upper wall and the topmost layer of the lower wall, while the lower wall (except for the topmost layer) is more preferentially wetted by the lubricant fluid rather than the working fluid; thus, a stable LLI can be obtained over LIS (this will be described in the next subsection). Besides, the lower wall is kept stationary, while the upper wall moves with a constant driven velocity (U_x) in the +x-direction to generate a transverse shear flow. Note that the driven velocity is varied to provide different shear rates on LIS, and thus the shear evolution of LLI is investigated. Note that periodic boundary conditions are applied along the x- and z-directions to simulate a periodically grooved LIS.

MDPD Parameters. In the current MDPD scheme, the interaction parameters are given in Table 1 in the DPD units. The subscripts w and i denote the working fluid in the channel and the infused liquid in the groove, respectively. In addition, subscripts l and s denote the liquid (both the working fluid and the lubricant fluid) and the solid wall, respectively.

Table 1. Parameters in DPD Units Used in the Present Study

parameter	symbol	value
particle density of the working fluid	ρ_w	6.70
particle density of the lubricant fluid	ρ_l	3.24
repulsion coefficient	$B_{ll} = B_{sl} = B_{ss}$	25.00
cut-off radius of attractive force	r_c	1.00
cut-off radius of repulsive force	r_d	0.75
friction coefficient	$\gamma_{sl} = \gamma_{ss}$	18.00
system temperature	k_BT	1.00
time step	Δt	0.01

In the current MDPD scheme, the interaction intensity between the solid wall and the fluid is primarily controlled by the attraction coefficient A. Note that a lower value of A means a relatively more attractive interaction between two particles, while a higher value of A means a relatively less attractive, i.e., repulsive, interaction between two particles. To effectively trap the lubricant within the surface grooves and form a stable LLI, the values of A between the topmost layer of the lower wall and the working fluid, the topmost layer of the lower wall and the lubricant fluid, the lower wall (except for the topmost layer) and the lubricant fluid, the working fluid, and the lubricant fluid are, respectively, set as −40.0, −4.0, −40.0, and 4.0. Thus, the topmost layer of the lower wall is hydrophilic to the working fluid while the lower wall (except for the topmost layer) is hydrophobic to the working fluid. On the other hand, the topmost layer of the lower wall is hydrophobic to the lubricant fluid while the lower wall (except for the topmost layer) is hydrophilic to the lubricant fluid. Especially, this coefficient is 0.0 between the fluid particles of the lubricant to simulate a light fluid, while it is −40.0 between the particles of the working fluid to simulate a relatively heavy fluid.

In an MDPD scheme, the mapping relations between the physical and DPD units for the length, mass, and time are $l^{\text{physical}} = [\mathrm{L}] \times l^{\text{DPD}}$, $m^{\text{physical}} = [\mathrm{M}] \times m^{\text{DPD}}$, and $t^{\text{physical}} = [\mathrm{T}] \times t^{\text{DPD}}$, respectively. Here, quantities in square brackets denote the scaling relationship between

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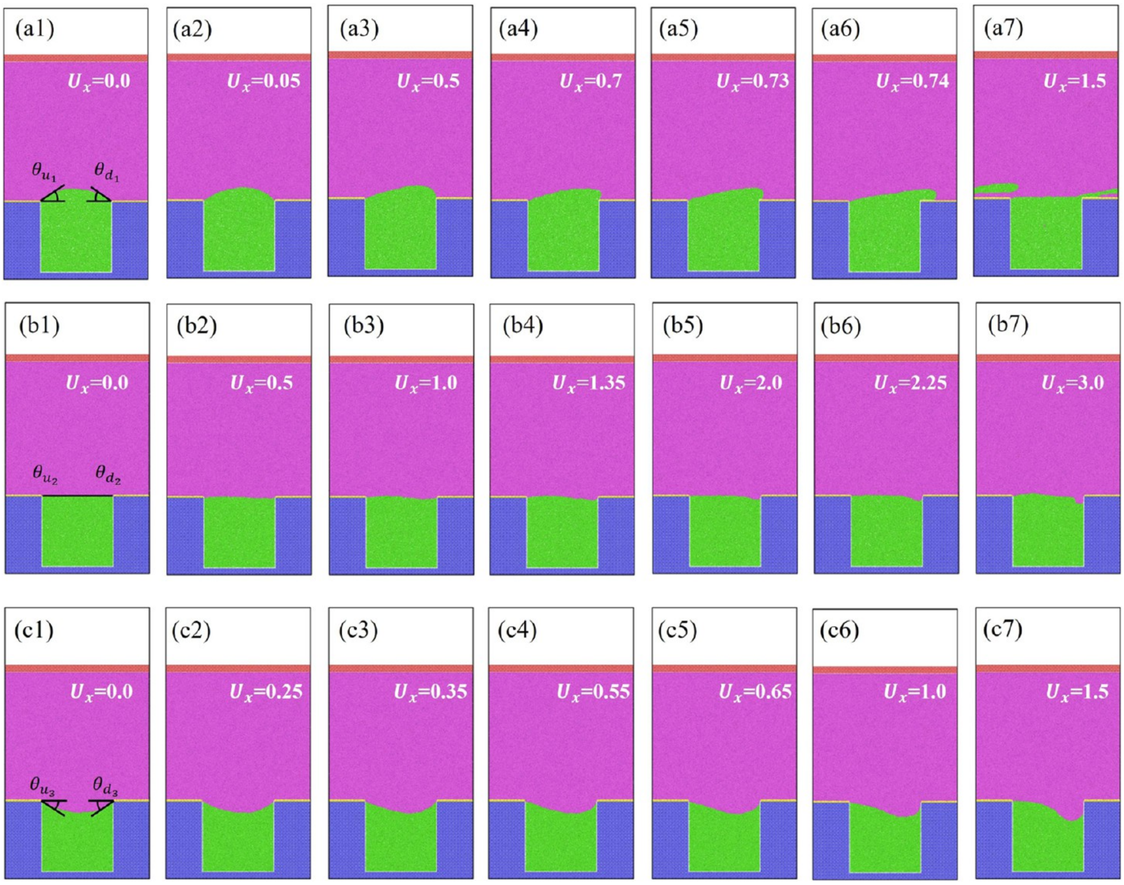

Figure 2. Shear evolution of LLIs with increasing driven velocity U_x for LLIs among different deflections θ. In (a−c), the deflections of LLIs are $\theta_1 = (\theta_{u_1} + \theta_{d_1})/2 = 37.0°$, $\theta_2 = (\theta_{u_2} + \theta_{d_2})/2 = 0.0°$, and $\theta_3 = (\theta_{u_3} + \theta_{d_3})/2 = -32.3°$, respectively. From (·1) to (·7), the driven velocity U_x gradually increases, and the value is indicated in the subfigures as well.

physical and DPD quantities, which are based on the density ρ, kinematic viscosity ν, and surface tension Γ of the current T-MDPD fluid and the physical fluid. Then, the dimensional analysis yields to, respectively

$$[\mathrm{M}] = [\mathrm{L}]^3 \frac{\rho^{\mathrm{DPD}}}{\rho^{\mathrm{physical}}} \tag{7}$$

$$[\mathrm{T}] = \left([\mathrm{M}]\frac{\Gamma^{\mathrm{physical}}}{\Gamma^{\mathrm{DPD}}}\right)^{0.5} \tag{8}$$

$$\frac{[\mathrm{L}]^2}{[\mathrm{T}]} = \frac{\nu^{\mathrm{DPD}}}{\nu^{\mathrm{physical}}} \tag{9}$$

As the friction coefficient of the lubricant varies in the current study to adjust the viscosity, take a classical simulation case, where the friction coefficient of the working fluid is 50.0 and it is 2.0 for the lubricant fluid, as an example to find the mapping relations. By applying the Poiseuille flow method, the dynamic viscosities of these two kinds of fluids are obtained as 74.31 and 1.22 in the DPD units. Based on the Irving−Kirkwood equation, the surface tension between these two kinds of fluids is acquired as 12.76 in the DPD units. Then, coupled with the density, viscosity, and surface tension of the current T-MDPD fluid in the channel with water, the scaling factors for the length, mass, and time are $[\mathrm{L}] = 1.416 \times 10^{-8}$ m, $[\mathrm{M}] = 4.229 \times 10^{-22}$ kg, and $[\mathrm{T}] = 2.331 \times 10^{-10}$ s, respectively. Note that while the viscosity of the lubricant fluid is determined by the friction coefficient, the density remains constant during this variation.

With a time step Δt of 0.01 in the DPD units, all simulations are carried out using the open-source MD code LAMMPS. For a given simulation input, the number of time steps for the initial equilibration is 5.0×10^5. After equilibration, the transverse shear flow is generated and developed to a steady state over 1.0×10^6 steps. Then, the entire simulation box is divided into narrow bins, with $\Delta x = 1.0$ in the DPD units and $\Delta y = 1.0$ in the DPD units parallel to the z-direction, and the density and velocity fields of the fluid in both the channel and the groove are computed, which are typically averaged over 1.0×10^6 time steps. Specifically, the density profile is used to acquire the position of LLI and the velocity profile is used to calculate the slip characteristics over LIS.

■ RESULTS AND DISCUSSION

Shear Evolution of LLIs. As indicated before, the deflection of the liquid−liquid interface (LLI) was adjusted by slowly tuning the channel height, and thus the contact angle of LLI at the groove edge was controlled as well. Specifically, the contact angle at the upstream groove edge is named as upstream contact angle θ_u, the contact angle at the downstream groove edge is named as downstream contact angle θ_d, and they satisfy the relation of $\theta_u = \theta_d$ after the initial equilibration without shear action. If LLI is protruded into the channel, a convex LLI will be obtained, and the corresponding θ_u and θ_d are both positive, while θ_u and θ_d are both negative for a concave LLI if LLI is protruded into the groove. And a flat LLI, where $\theta_u = \theta_d = 0.0°$, is defined, while the interface is parallel to the topmost layer of the lower wall. Here, an average contact angle $\theta(=\theta_u + \theta_d)/2$ is used to illustrate the deflection of an LLI without shear action. Then, the shear evolution of LLIs with increasing driven velocity acted over LLIs (U_x) for the three different deflections, i.e., a convex LLI ($\theta_1 = (\theta_{u_1} + \theta_{d_1})/2 = 37.0°$), a flat LLI ($\theta_2 = (\theta_{u_2} + \theta_{d_2})/2 =$

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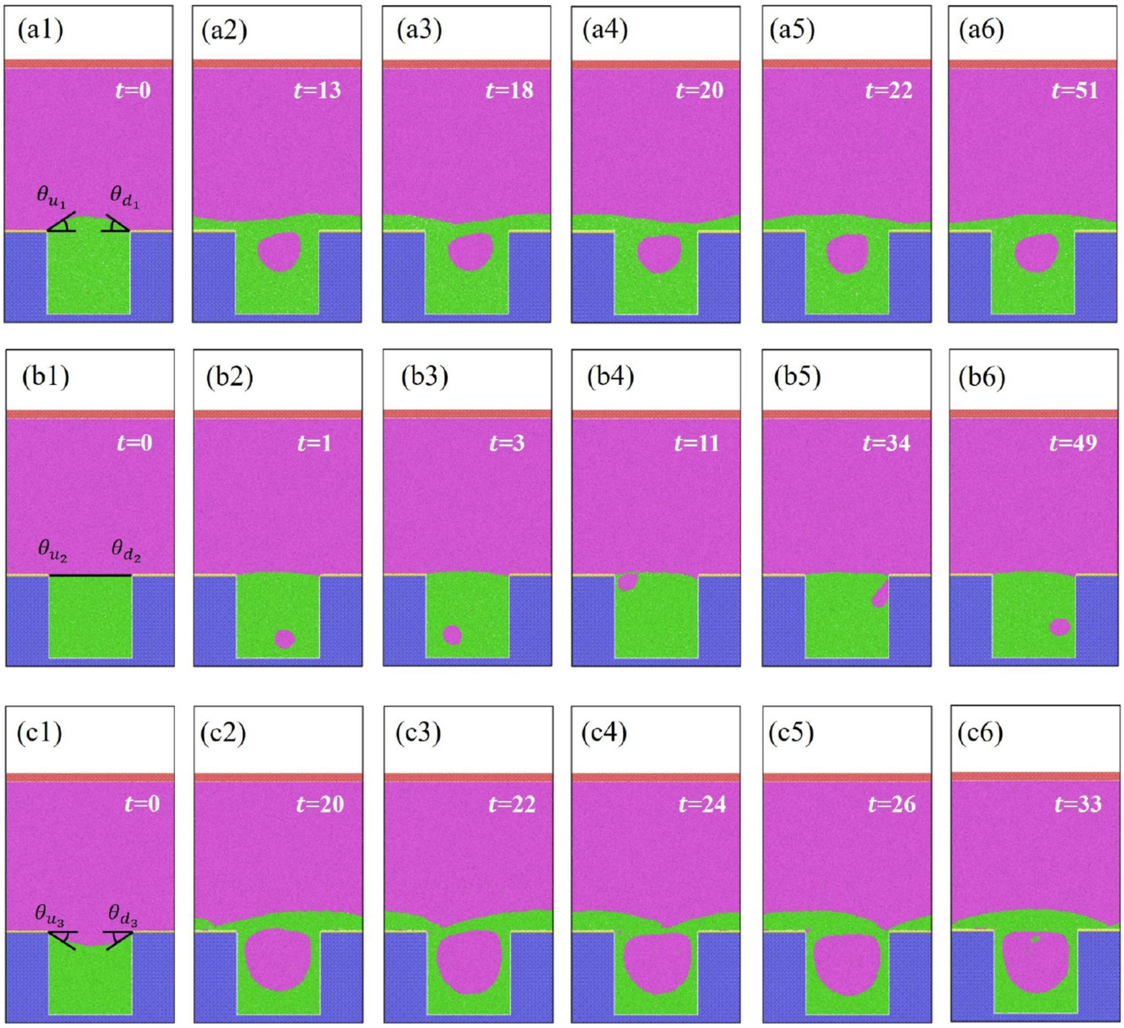

Figure 3. Evolution of LLIs with increasing shear time t for LLIs among different deflections θ ($U_x = 6.0$). In (a−c), the deflections of LLIs are $\theta_1 = (\theta_{u_1} + \theta_{d_1})/2 = 37.0°$, $\theta_2 = (\theta_{u_2} + \theta_{d_2})/2 = 0.0°$, and $\theta_3 = (\theta_{u_3} + \theta_{d_3})/2 = -32.3°$, respectively. From (·1) to (·6), the shear time t gradually increases and the value is indicated in the subfigures as well.

0.0°), and a concave LLI ($\theta_3 = (\theta_{u_3} + \theta_{d_3})/2 = -32.3°$), is first investigated.

For a convex LLI as shown in Figure 2(a), it is found that while the driven velocity U_x is relatively lower, LLI is relatively steady and symmetric as θ_u is almost equal to θ_d, and both the upstream and the downstream liquid−liquid−solid contact lines pin at groove edges. While U_x gradually increases, LLI becomes unsteady, and it deforms apparently. Specifically, with a decrease of θ_u and an increase of θ_d, LLI gradually deforms, while the three phase contact lines still pin at groove edges. With a further increase of U_x, the downstream three phase contact line first depins, while the upstream contact line still pins at the groove edge of the lower wall. Then, with a much further increase of U_x, the upstream three phase contact line depins as well, and thus a layer of lubricant tends to be formed over the grooved surface. Note that the liquid−liquid interface bulges slightly on top of the solid ridge but overall stays rather flat along the grooved surface.

Different from the evolution of a convex LLI under the action of a shear flow, the evolution of a flat LLI is shown in Figure 2(b). With an increase of the driven velocity U_x, the upstream interface tends to protrude into the channel while the downstream interface tends to protrude into the groove, thus a positive θ_u and a negative θ_d are obtained. And both the upstream and downstream liquid−liquid−solid contact lines pin at groove edges. Then, as U_x continues to grow, the downstream three phase contact line depins and begins to move down along the groove side wall, while the upstream contact line does not move. On the other hand, the shear evolution of a concave LLI is shown in Figure 2(c). Results show that with increasing U_x, θ_u gradually increases (i.e., a decrease of the absolute value of θ_u) while θ_d gradually decreases (i.e., an increase of the absolute value of θ_d), and eventually the downstream three phase contact line moves down along the groove side wall while the upstream contact line does not move.

Meanwhile, taking account of the three LLIs with different deflections as shown in Figure 2, the evolution of LLIs with increasing shear time under a severe shear action, i.e., a much higher value of the driven velocity U_x, is shown in Figure 3. Results show that the working fluid will roll into the lubricant fluid, and thus both the working fluid and the lubricant fluid will be infused within the groove, which is definitely different from the standard design of an LIS where only the lubricant fluid is infused within the surface structures and thus LIS is degraded. Note that the flow inside gradually becomes unsteady as the rolled working fluid gradually moves but always moves inside the groove.

As shown above, LLI over LISs will gradually become unstable under the action of a flow shear. Here, the force balance of a convex LLI is schematically shown in Figure 4 to explore the

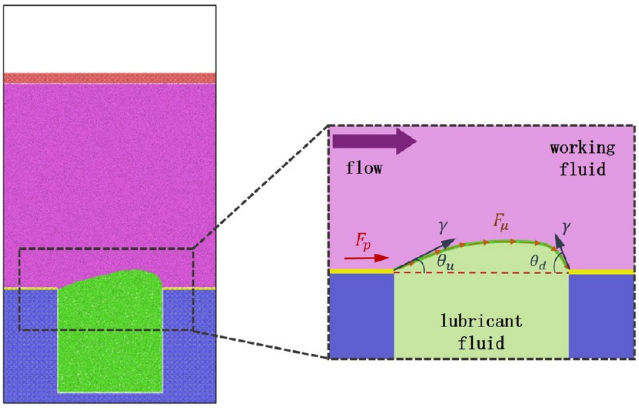

Figure 4. Schematic shows the forces acting on a convex LLI.

evolution process of LLI. There are three forces, the capillary force F_Γ induced by the surface tension, the shear force F_μ induced by the flow, and the pressure force F_p induced by the pressure difference, which act on LLI. Since a boundary condition is applied along the z-direction, the protrusion height of the interface along the y-direction is much smaller than the interface width along the z-direction, and F_p has a negligible effect on LLI and thus the balance between F_Γ and F_μ governs the interface.[50] Specifically, F_Γ can be obtained by $F_\Gamma = \Gamma L_z(\cos\theta_u - \cos\theta_d)$, where Γ is the surface tension between the working fluid and the lubricant fluid, and L_z is the length of the interface along the z-direction. F_μ can be qualitatively determined by $F_\mu \sim \mu_w \dot{\gamma} L_z L_x$, where μ_w is the dynamic viscosity of the working fluid, $\dot{\gamma}$ represents an average shear rate acting on the interface and is positively related to the driven velocity U_x, and L_x is the length of the interface along the x-direction.

With an increase of the driven velocity U_x, not only LLI becomes to deform as shown in Figure 2(a), but the local shear rate along the interface also increases as shown in Figure 5, which thus provides a larger average shear rate, i.e., a larger shear force F_μ, at the interface. Specifically, as the upstream contact angle of the interface θ_u gradually goes down and the downstream contact angle of the interface θ_d gradually goes up, the capillary force F_Γ tends to be larger to resist increasing F_μ, and thus a balance between F_Γ and F_μ is acquired again. Once F_Γ induced by the interface deformation is not able to balance F_μ induced by the flow, the liquid−liquid−solid three phase contact line will depin from the groove edge, and the interface will eventually be destroyed, which will thus make LISs lose their initial functions or advantages. Specifically, for a convex LLI, the critical destroy state, i.e., where the downstream three phase contact line begins to depin, corresponds to the value of θ_d increasing with the advancing angle of the lubricant on the topmost hydrophilic layer of the lower wall. Note that the lower wall, except for the topmost hydrophilic layer, is set as hydrophobic to prevent the wetting of the working liquid. Similarly, the force balance acting on a flat LLI or a concave LLI is also dominated by F_Γ and F_μ. However, for such cases, the downstream three phase contact line begins to move down along the groove side wall once the value of θ_d reaches the receding

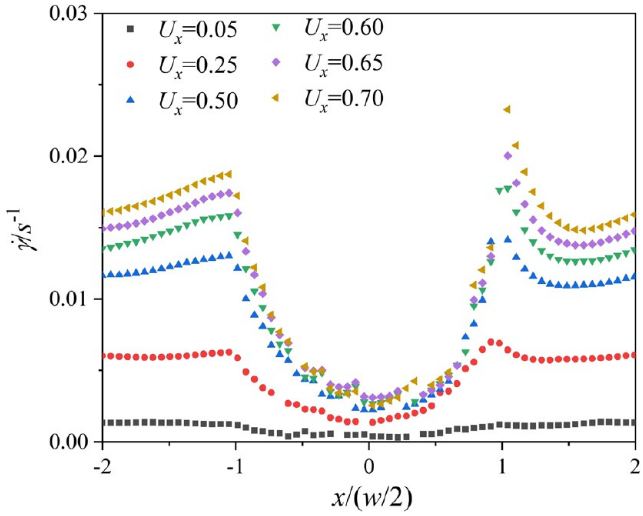

Figure 5. Variation of the local shear rate $\dot{\gamma}$ along LLI with increasing driven velocity U_x. The interface deflection is $\theta = 37.0°$, and the origin is located at the center of the interface.

angle of the lubricant on the lower hydrophobic wall (except for the topmost layer of the lower hydrophilic wall), while the upstream contact line does not move.

Critical Failure of LLIs. Note that for a given shear driven velocity U_x acted on LIS, the deformation degree of LLI is different as shown in Figure 2. For example, when $U_x = 0.5$ in the DPD units, the convex LLI deforms obviously, while a flat LLI nearly has no change. Then, when U_x increases to a value of 1.0 in the DPD units, the flat LLI only deforms a little while the downstream three phase contact line has depinned and moved for a concave LLI. Here, the critical failure shear rate $\dot{\gamma}_c$ ($=U_{x,c}/h$, where $U_{x,c}$ is the critical shear velocity, and h is the height of the channel) for various LLIs with different deflections is identified in Figure 6. Specifically, the critical failure of LLI corresponds to that the downstream three phase contact line depins and begins to move of LLI over an LIS. It can be found that with an increase of θ, $\dot{\gamma}_c$, which is plotted as red half-solid symbols, increases first and then decreases, and the maximum value of $\dot{\gamma}_c$ corresponds to

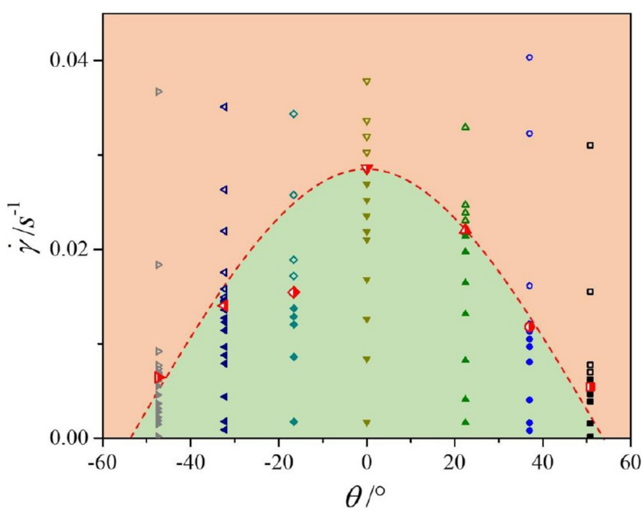

Figure 6. Variation of critical failure shear rate $\dot{\gamma}_c$ for LLIs with increasing deflections θ. The solid symbols, hollow symbols, and half-solid symbols represent the steady, failure, and critical failure modes of LLIs, respectively.

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Langmuir 2025, 41, 5158−5169

a flat LLI. Specifically, for convex LLIs, while θ increases from 37.0° to a value of 50.8°, $\dot{\gamma}_c$ decreases from 0.012 s^{-1} to a value of 0.0054 s^{-1}. When LLI is concave, $\dot{\gamma}_c$ decreases from 0.014 to 0.0064 s^{-1} if θ decreases from −32.3° to a value of −47.36°. This does mean that an increase of the protrusion deflection either to the channel or to the groove will make LLI less stable under the action of a flow shear, i.e., the flat LLI is more stable under the same shear action.

Then, both the viscosity ratio (i.e., the dynamic viscosity ratio between the working fluid to that of the lubricant fluid) and the fraction (i.e, the ratio of the groove width to the simulation domain length in the x-direction) of LLIs are varied to investigate the effect of LLI parameters on shear evolution of LLI over LISs. First, the effect of the viscosity ratio N on the evolution of LLI is explored. Specifically, by varying the friction coefficient of the lubricant fluid while the friction coefficient is constantly set at 50.0 in the DPD units for the working fluid, a series of values of N are obtained as shown in Table 2. Note that a superhydrophobic surface (SHS) is also taken into account to provide a higher value of N. Specifically, by deleting the particles infused within surface structures, an SHS is thus achieved and a classical value of N = 55.65 is obtained as well.

Table 2. Variation of the Viscosity Ratio N in the Simulations

friction coefficient	viscosity	viscosity ratio
4.5	1.65	45.02
8	2.29	32.45
18	4.22	17.62
40.5	8.50	8.74

As it is found that a flat LLI is relatively stable based on above simulations, the flat interfaces with different values of N are taken as examples. From Figure 7, it can be seen that as the shear driven velocity U_x gradually increases, the evolutions of LLIs with different values of N are similar while the downstream liquid−liquid−solid three phase contact lines gradually depin and then move along the side wall of grooves. While the lubricant fluid becomes more viscous, i.e., N decreases, the interface tends to be more stable under the same shear action. For example, while U_x = 1.0 in the DPD units, LLI with a value of N_3 = 8.74 nearly keeps flat, while LLI with a value of N_2 = 17.62 and a value of N_1 = 45.02 both deform and the latter deforms severely.

Then, the effect of the LLI fraction α (which corresponds to the groove width w) is explored as shown in Figure 8. Results show that with an increase of the driven velocity U_x, the interface gradually deforms while the upstream interface tends to protrude into the channel and the downstream interface tends to protrude into the groove, then the downstream liquid−liquid−solid three phase contact line depins from the groove edge and moves along the groove side wall with a further increase of U_x. With an expansion of the groove, the deform degree of the interface performs differently. For example, while U_x = 1.0 in the DPD units, LLI over a narrow groove with the

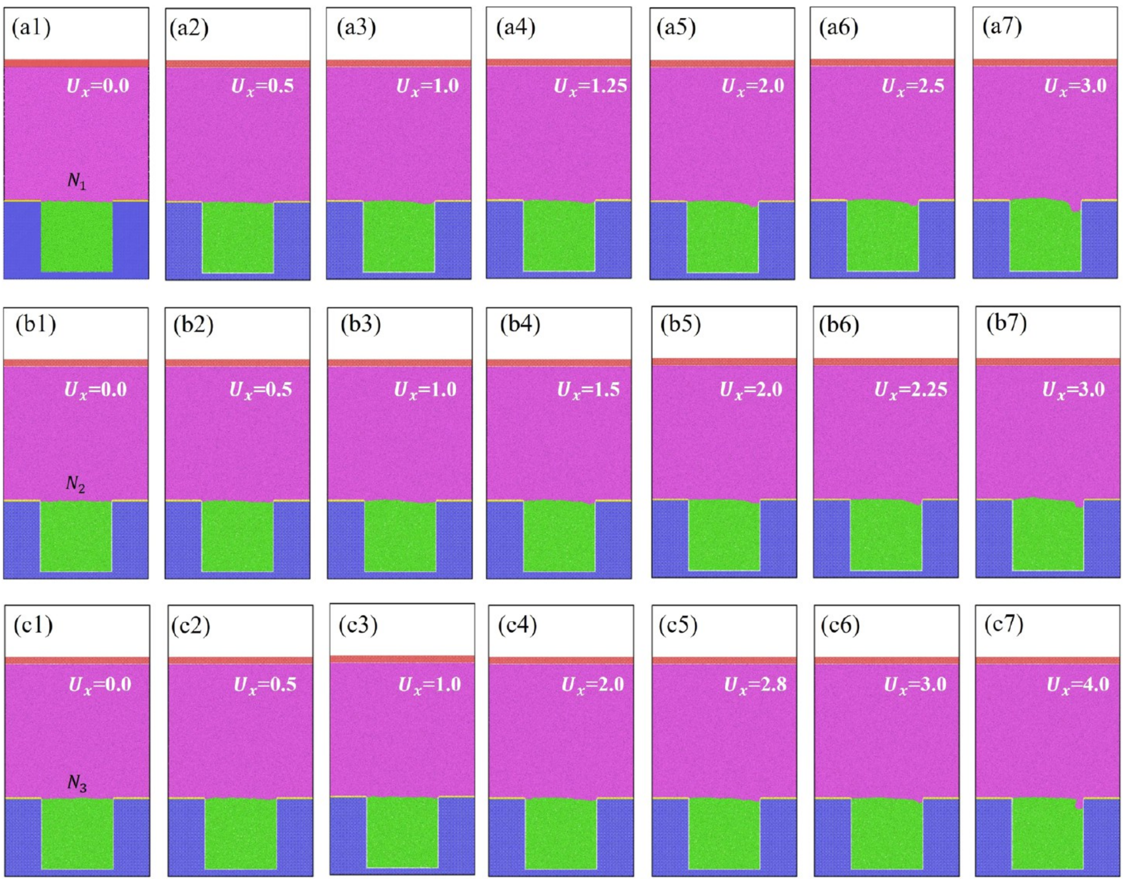

Figure 7. Shear evolution of LLIs with increasing driven velocity U_x for LLIs among different viscosity ratios N. In (a−c), the viscosity ratios of LLIs are N_1 = 32.45, N_2 = 17.62, and N_3 = 8.74, respectively. From (·1) to (·7), the driven velocity U_x gradually increases and the value is indicated in the subfigures as well.

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Langmuir 2025, 41, 5158−5169

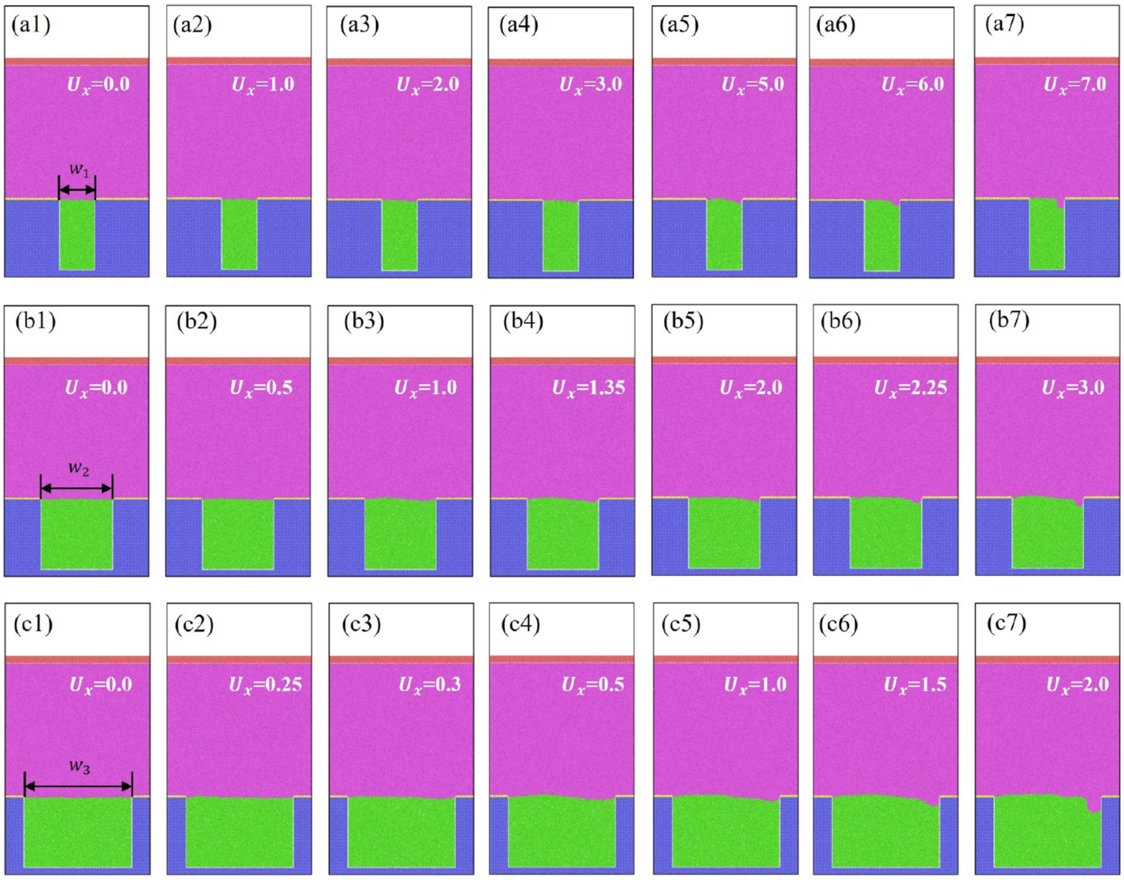

Figure 8. Shear evolution of LLIs with increasing driven velocity U_x for LLIs among different groove widths w. In (a−c), the groove widths of LLIs are w_1 = 15.60, w_2 = 31.55, and w_3 = 47.50, respectively. From (·1) to (·7), the driven velocity U_x gradually increases and the value is indicated in the subfigures as well.

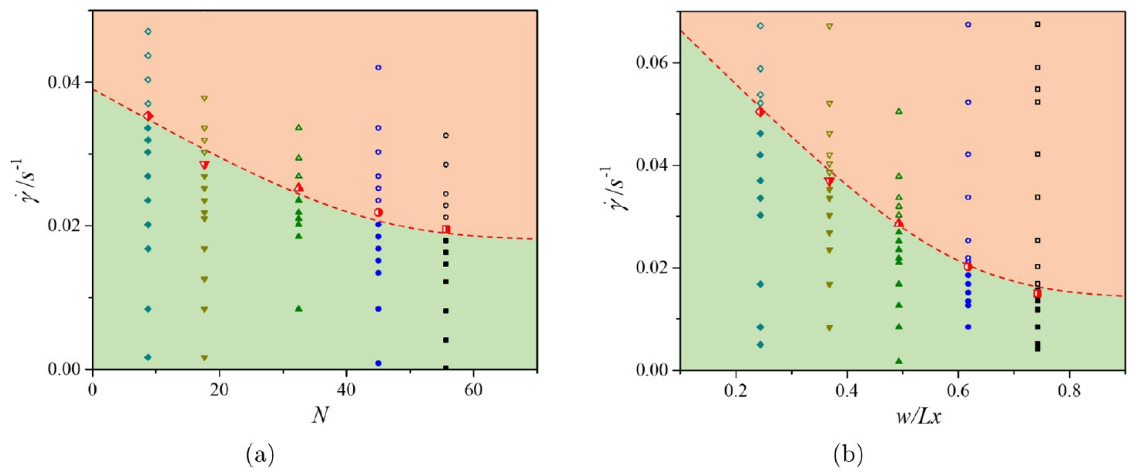

Figure 9. Variation of critical failure shear rate $\dot{\gamma}_c$ with increasing (a) viscosity ratio N and (b) LLI fraction α. The solid symbols, hollow symbols, and half-solid symbols represent the steady, failure, and critical failure modes of LLIs, respectively.

width w_1 = 15.60 in the DPD units nearly keeps flat, while LLI over a wider groove with the width w_2 = 31.55 in the DPD units begins to deform, and LLI over the widest groove with the width w_3 = 47.50 in the DPD units deforms apparently.

Besides, the critical failure shear rate $\dot{\gamma}_c$ of LLIs with increasing viscosity ratio N is plotted as red symbols in Figure 9(a). Consistent with the evolution of LLI, $\dot{\gamma}_c$ gradually decreases with increasing N. Meanwhile, the variation of $\dot{\gamma}_c$ for LLIs with increasing LLI fraction α is explored in Figure 9(b). Results show that $\dot{\gamma}_c$ gradually decreases with increasing α. Whereas a higher value of N or a larger value of α is beneficial to promote the slip at LLI,[20−22] an LLI with a higher value of N or a larger value of α is less stable, which means that both the slippage and the stability of LLI should be considered or balanced when designing an effective and robust LIS.

Effective Slippage of LLIs. In this section, the slip characteristic of LIS is analyzed to evaluate the shear deformation of LLI over an LIS. While the interface is always

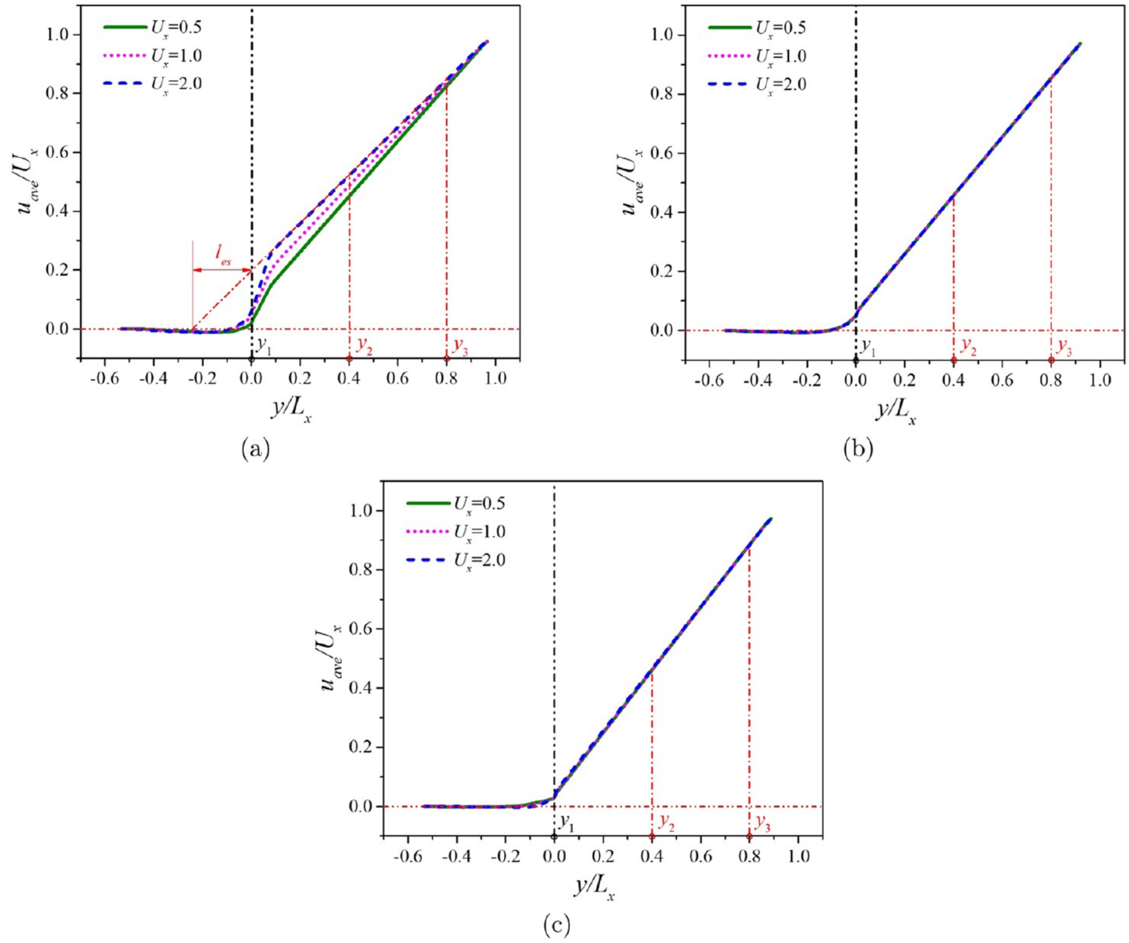

Figure 10. Normal profiles of average velocity u_{ave} with different driven velocities U_x for various LLI deflections θ. (a) $\theta_1 = 37.0°$, (b) $\theta_2 = 0.0°$, and (c) $\theta_3 = -32.3°$. y_1 denotes the position of LLI when calculating l_{es}, and y_2 and y_3 denote the linearly fitting region of the velocity.

irregular, the effective slip length, not the local slip length, is used to characterize the flow features over an LIS. Note that the effective slip length l_{es} is the equivalent slip required on a smooth surface that would produce the same flow conditions far away from the composite surface, i.e., l_{es} is an averaged quantity equal to the distance below the surface at which the velocity would extrapolate to zero. Specifically, l_{es} was directly measured as follows: the normal profiles of the streamwise velocity (in the x-direction) in the bulk region were first averaged along the entire channel, the averaged velocity profile was then linearly extrapolated to the location of LLI to determine the effective slip velocity and the velocity gradient, and thus l_{es} was calculated based on the Naiver slip model. In this model, $l_{\mathrm{es}} = u_{\mathrm{es}}/\dot{\gamma}$, where $\dot{\gamma} = \mathrm{d}u/\mathrm{d}y$ represents the shear rate, i.e., the normal gradient of the velocity near the interface, and u_{es} is the effective slip velocity at the interface.

Here, the profiles of the average velocity u_{ave} along the y-direction with different driven velocities for various LLI deflections are first compared in Figure 10. While LLI is convex (with a deflection of $\theta_1 = 37.0°$), the velocity profiles are quite different among different driven velocities as shown in Figure 10(a). Specifically, with an increasing driven velocity, the velocity profile tends to be higher as well, which means that shear evolution induces a stronger modification of the flow in the bulk region. When LLI is flat (with a deflection of $\theta_2 = 0.0°$) or concave (with a deflection of $\theta_3 = -32.3°$), the velocity profiles are almost similar as the driven velocity varies, which corresponds to a relatively small modification of the flow.

Then, the effective slip lengths l_{es} among a series of velocities for various LLIs with different deflections are compared in Figure 11. Note that the LLI deflections are different and they all gradually become irregular with increasing driven velocity, and the location of LLI is assumed and united to be the position of the topmost layer of the lower wall, which is indicated as y_1 in Figure 10. Besides, the linearly fitting regions in the bulk region are indicated as y_2 to y_3 in Figure 10, and the calculation of l_{es} is also schematically shown in Figure 10(a). From Figure 11, results show that when the shear rate $\dot{\gamma}$ is relatively low, l_{es} is relatively small and almost constant for the three interfaces, which indicates an independence on the shear rate, and the only effect of the flow is to deform LLI slightly. Then, with an increase of $\dot{\gamma}$, l_{es} almost remains constant for a flat LLI, while that of a convex or a concave LLI changes apparently. Specifically, for a convex LLI, l_{es} increases sharply at a certain $\dot{\gamma}$. According to the evolution of a convex LLI, the sharp increase, i.e., the enhancement of l_{es}, is related to the depinning of the liquid−liquid−solid three phase contact line, and thus the fraction of the lower wall exposed to the working fluid decreases and more working fluid can slide on top of the lubricant fluid to promote slippage. For a concave LLI, l_{es} increases continuously with a further increase of $\dot{\gamma}$, while the increase rate is relatively much lower than that of a convex LLI as the evolution of the interface is less severe.

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Langmuir 2025, 41, 5158−5169

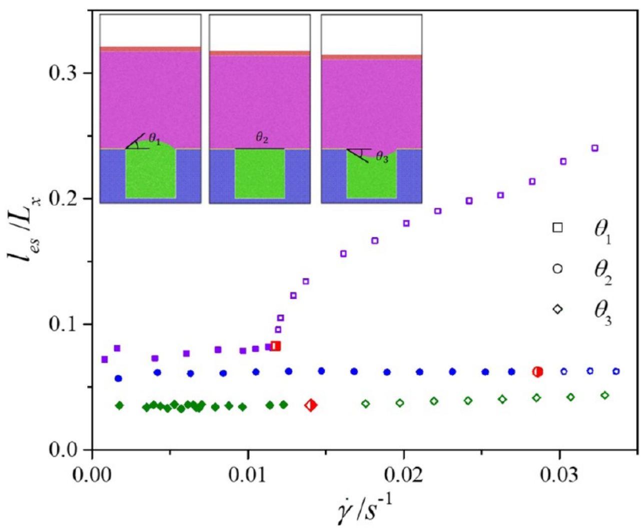

Figure 11. Variation of effective slip length l_{es} with increasing shear rate $\dot{\gamma}$ for various LLIs among different deflections θ. $\theta_1 = 37.0°$, $\theta_2 = 0.0°$, and $\theta_3 = -32.3°$. The solid symbols, hollow symbols, and half-solid symbols represent the steady, failure, and critical failure modes of LLIs, respectively.

Actually, the variation of the effective slip length for the convex LLI over LISs with increasing shear rate is similar to that of an SHS exposed to a shear flow.[50] Specifically, by numerically simulating a shear flow over a periodically patterned substrate with entrapped concave gas bubbles, Gao et al. found that the gas−liquid−solid three phase contact lines can be pinned, depinned, or eliminated depending on the competition between the shear force and the surface tension and indicated that a significantly enhanced effective slip length can be obtained when the shear rate is larger and the bubbles are transformed into a continuous gas film. In the current study, the effect of the LLI deflection on the shear evolution and the slippage are also included. Results show that the effective slippage is not only related to the shear rate but also influenced by the interface deflection. Although the slippage is the highest for a convex LLI, the flat LLI performs stablest under the same shear action, so both the shear resistance and the slippage of LLI should be considered when designing an effective LIS.

■ CONCLUSIONS

The structured liquid-infused surfaces (LISs), with an immiscible lubricant fluid infused within the structures, are a promising way to reduce the flow drag, and the liquid−liquid interface (LLI) is the key to cause flow slippage and thus promote a drag-reducing effect. Although LISs can avoid the deleterious effects of pressure fluctuations and gas dissolution that arise with gas-trapped superhydrophobic surfaces (SHSs), they do suffer from the action of flow shear and thus make LLI unstable. In the current study, whether the interface protruded inside or outside the structures, the newly proposed transverse many-body dissipative dynamics simulation method was applied to explore the shear evolution of LLI, as well as the corresponding slip properties were analyzed with a transverse Couette flow over a periodically grooved LIS.

Results show that with an increase of the shear rate, LLI gradually deforms to produce a larger capillary force to resist the shear force. Once the capillary force is unable to balance the shear force, the downstream three phase contact line depins and begins to move, which eventually makes LIS lose its original design. Compared with a convex LLI or a concave LLI, a flat LLI is relatively stable under the same shear action. And a decrease of the viscosity ratio and the LLI fraction can both strengthen the shear resistance of an LLI. Note that when the shear rate is relatively lower, the effective slippage is less sensitive to the shear rate, i.e., the effective slip length is independent of the shear rate and the only effect of the flow is to deform LLI slightly. Then, with a further increase of the shear rate, as the contact line depinning occurs and a lubricant layer tends to form for a convex LLI, the infused lubricant greatly enhances apparent slippage, while the slippage cannot be obviously enhanced for a flat or a concave LLI. The value of the effective slippage is influenced by the interface deflection, as the slippage is the highest for a convex LLI, then a flat LLI, and the slippage of a concave LLI is the lowest. The current results not only indicate that the slippage is related to the interface deflection and shear rate but also suggest that the shear resistance and the slippage of LLI should be both considered when designing a relatively robust LIS to reduce the flow drag.

■ AUTHOR INFORMATION

Corresponding Author

Haibao Hu − *School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China;* orcid.org/0000-0001-5201-2001; Email: huhaibao@nwpu.edu.cn

Authors

Liuzhen Ren − *School of Construction Machinery, Chang'an University, Xi'an, Shaanxi 710064, People's Republic of China*

Jiangzhuo Ren − *School of Construction Machinery, Chang'an University, Xi'an, Shaanxi 710064, People's Republic of China*

Luyao Bao − *State Key Laboratory of Solid Lubrication, Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences, Lanzhou, Gansu 730000, People's Republic of China*

Jun Wen − *School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China*

Min Ye − *School of Construction Machinery, Chang'an University, Xi'an, Shaanxi 710064, People's Republic of China;* orcid.org/0000-0002-8301-5843

Complete contact information is available at:
https://pubs.acs.org/10.1021/acs.langmuir.4c04428

Notes

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■ REFERENCES

(1) Perlin, M.; Dowling, D.; Ceccio, S. Freeman scholar review: Passive and active skin-friction drag reduction in turbulent boundary layers. *J. Fluid Eng.* **2016**, *138*, No. 091104.

(2) Tian, G. Z.; Zhang, Y. S.; Feng, X. M.; Hu, Y. S. Focus on bioinspired textured surfaces toward fluid drag reduction: Recent progresses and challenges. *Adv. Eng. Mater.* **2022**, *24*, No. 2100696.
(3) Wong, T. S.; Kang, S. H.; Tang, S.; Smythe, E. J.; Hatton, B. D.; Grinthal, A.; Aizenberg, J. Bioinspired self-repairing slippery surfaces with pressure-stable omniphobicity. *Nature* **2011**, *477*, 443−447.
(4) Hardt, S.; McHale, G. Flow and drop transport along liquid-infused surfaces. *Annu. Rev. Fluid Mech.* **2022**, *54*, 83−104.
(5) Pakzad, H.; Nouri-Borujerdi, A.; Moosavi, A. Studying liquid-Iinfused surfaces with affordable surface features infiltrated with various lubricants from corrosion, biofouling, and fluid drag viewpoints. *Langmuir* **2023**, *39*, 10978−10992.
(6) Rothstein, J. P. Slip on superhydrophobic surfaces. *Annu. Rev. Fluid Mech.* **2010**, *42*, 89−109.
(7) Park, H.; Choi, C. H.; Kim, C. J. Superhydrophobic drag reduction in turbulent flows: a critical review. *Exp. Fluids* **2021**, *62*, No. 229.
(8) Wang, T.; Wang, Z. K. Liquid-repellent surfaces. *Langmuir* **2022**, *38*, 9073−9084.
(9) Huang, S. L.; Lv, P. Y.; Duan, H. L. Morphology evolution of liquid−gas interface on submerged solid structured surfaces. *Extreme Mech. Lett.* **2019**, *27*, 34−51.
(10) Choi, W.; Kang, M.; Park, J. Y.; Jeong, H. E.; Lee, S. J. Enhanced air stability of superhydrophobic surfaces with flexible overhangs of re-entrant structures. *Phys. Fluids* **2021**, *33*, No. 022001.
(11) Kim, H.; Park, H. Diffusion characteristics of air pockets on hydrophobic surfaces in channel flow: Three-dimensional measurement of air-water interface. *Phys. Rev. Fluids* **2019**, *4*, No. 074001.
(12) Fu, M. K.; Arenas, I.; Leonardi, S.; Hultmark, M. Liquid-infused surfaces as a passive method of turbulent drag reduction. *J. Fluid Mech.* **2017**, *824*, 688−700.
(13) Arenas, I.; García, E.; Fu, M. K.; Orlandi, P.; Hultmark, M.; Leonardi, S. Comparison between super-hydrophobic, liquid infused and rough surfaces: A direct numerical simulation study. *J. Fluid Mech.* **2019**, *869*, 500−525.
(14) Vega-Sánchez, C.; Neto, C. Fluid slip and drag reduction on liquid-infused surfaces under high static pressure. *Langmuir* **2024**, *40*, 4460−4467.
(15) Lee, J.; Lee, M. H.; Choi, C. H. Design of robust lubricant-infused surfaces for anti-corrosion. *ACS Appl. Mater. Interfaces* **2022**, *14*, 2411−2423.
(16) Subramanyam, S. B.; Rykaczewski, K.; Varanasi, K. K. Ice adhesion on lubricant-impregnated textured surfaces. *Langmuir* **2013**, *29*, 13414−13418.
(17) Guo, Z. Q.; Monga, D.; Shan, L.; Boylan, D.; Dai, X. M. Coarsening-induced disappearing droplets contribute to condensation. *Droplet* **2022**, *1*, 170−181.
(18) Sun, P. C.; Hao, X. Q.; Jin, Y. K.; Yin, Y. Y.; Wu, C. Y.; Zhang, J.; Gao, L. J.; Wang, S.; Wang, Z. K. Heterogenous slippery surfaces: enabling spontaneous and rapid transport of viscous liquids with viscosities exceeding 10000 mPa s. *Small* **2023**, *19*, No. 2304218.
(19) Yan, X. T.; Song, Y. X.; Zheng, H. X.; Cui, H. Z.; Wang, Z. K.; Xu, W. H. A bubble energy generator featuring lubricant-impregnated surface with high durability and efficiency. *Nano Energy* **2024**, *121*, No. 109238.
(20) Solomon, B. R.; Khalil, K. S.; Varanasi, K. K. Drag Reduction using Lubricant-Impregnated Surfaces in Viscous Laminar Flow. *Langmuir* **2014**, *30*, 10970.
(21) Rosenberg, B. J.; Buren, T. V.; Fu, M. K.; Smits, A. J. Turbulent drag reduction over air- and liquid- impregnated surfaces. *Phys. Fluids* **2016**, *28*, No. 015103.
(22) Ren, L. Z.; Zhang, M. Z.; Wen, J.; Bao, L. Y.; Hu, H. B. Drag reduction with lubricant rings over a patterned liquid-infused cylinder. *Tribol. Int.* **2024**, *198*, No. 109870.
(23) Yan, H. L.; Qian, F.; Jiao, K.; Zhang, W. Y.; Tan, Z. T.; Zhao, L. R.; Wang, Q. W.; Zhao, C. L. Copper capillaries with lubricant-infused walls: fabrication and drag reduction performance. *Microfluid. Nanofluid.* **2022**, *26*, No. 77.
(24) Kim, H. N.; Jeong, S. W.; Baik, B. G.; Jang, H.; Jeong, H. E.; Lee, S. J. Bioinspired low-friction surface coating with lubricant-infused spherical cavities for sustainable drag reduction. *Appl. Surf. Sci.* **2023**, *628*, No. 157365.
(25) Lee, S. J.; Kim, H. N.; Choi, W.; Yoon, G. Y.; Seo, E. A nature-inspired lubricant-infused surface for sustainable drag reduction. *Soft Matter* **2019**, *15*, 8459−8467.
(26) Chen, H. S.; Gao, Y.; Stone, H. A.; Li, J. Fluid bearing" effect of enclosed liquids in grooves on drag reduction in microchannels. *Phys. Rev. Fluids* **2016**, *1*, No. 083904.
(27) Asmolov, E. S.; Nizkaya, T. V.; Vinogradova, O. I. Enhanced slip properties of lubricant-infused grooves. *Phys. Rev. E* **2018**, *98*, No. 033103.
(28) Chang, J.; Jung, T.; Choi, H.; Kim, J. Predictions of the effective slip length and drag reduction with a lubricated micro-groove surface in a turbulent channel flow. *J. Fluid Mech.* **2019**, *874*, 797−820.
(29) Ren, L. Z.; Hu, H. B.; Bao, L. Y.; Priezjev, N. V.; Wen, J.; Xie, L. Two local slip modes at the liquid-liquid interface over liquid-infused surfaces. *Phys. Fluids* **2022**, *34*, No. 082017.
(30) Scarratt, L. R. J.; Zhu, L.; Neto, C. Large effective slip on lubricated surfaces measured with colloidal probe AFM. *Langmuir* **2020**, *36*, 6033−6040.
(31) Kim, J. H.; Rothstein, J. P. Delayed lubricant depletion on liquid-infused randomly rough surfaces. *Exp. Fluids* **2016**, *57*, No. 81.
(32) Vega-Sánchez, C.; Neto, C. Slightly depleted lubricant-infused surfaces are no longer slippery. *Langmuir* **2022**, *38*, 10568−10574.
(33) Yan, H. L.; Zhang, W. Y.; Cui, Y. H.; Qian, F.; Wei, D. M.; Guo, P. P.; Jiao, K.; Huang, J.; Wang, Q. W.; Zhao, C. L. Durable drag reduction and anti-corrosion for liquid flows inside lubricant-infused aluminum/copper capillaries. *Chem. Eng. Sci.* **2023**, *266*, No. 118275.
(34) Wexler, J. S.; Jacobi, I.; Stone, H. A. Shear-driven failure of liquid-infused surfaces. *Phys. Rev. Lett.* **2015**, *114*, No. 168301.
(35) Jacobi, I.; Wexler, J. S.; Stone, H. A. Overflow cascades in liquid-infused substrates. *Phys. Fluids* **2015**, *27*, No. 082101.
(36) Liu, Y.; Wexler, J. S.; Schonecker, C.; Stone, H. A. Effect of viscosity ratio on the shear-driven failure of liquid-infused surfaces. *Phys. Rev. Fluids* **2016**, *1*, No. 074003.
(37) Mazor, L.; Stone, H. A.; Jacobi, I. Effect of streamwise cross-sectional variation on liquid retention in liquid-infused substrates under an external flow. *Phys. Rev. Fluids* **2019**, *4*, No. 074003.
(38) Fu, M. K.; Chen, T. H.; Arnold, C. B.; Hultmark, M. Experimental investigations of liquid-infused surface robustness under turbulent flow. *Exp. Fluids* **2019**, *60*, No. 100.
(39) Asmolov, E. S.; Nizkaya, T. V.; Vinogradova, O. I. Flow-driven collapse of lubricant-infused surfaces. *J. Fluid Mech.* **2020**, *901*, No. A34.
(40) Ahuja, R.; Gaddam, A.; Joshi, S. S.; Agrawal, A. Characterization of the liquid-lubricant interface in a dovetail cavity for a viscous laminar flow. *Ind. Eng. Chem. Res.* **2023**, *62*, 2944−2958.
(41) Warren, P. B. Vapor-liquid coexistence in many-body dissipative particle dynamics. *Phys. Rev. E* **2003**, *68*, No. 066702.
(42) Español, P.; Warren, P. B. Perspective: Dissipative particle dynamics. *J. Chem. Phys.* **2017**, *146*, No. 150901.
(43) Groot, R. D.; Warren, P. B. Dissipative particle dynamics: Bridging the gap between atomistic and mesoscopic simulation. *J. Chem. Phys.* **1997**, *107*, 4423−4435.
(44) Li, Z.; Bian, X.; Tang, Y.; Karniadakis, G. E. A dissipative particle dynamics method for arbitrarily complex geometries. *J. Comput. Phys.* **2018**, *355*, 534−547.
(45) Wang, C.; Wu, X.; Zhang, H.; Hao, P.; He, F.; Zhang, X. A many-body dissipative particle dynamics study of eccentric droplets impacting inclined fiber. *Phys. Fluids* **2021**, *33*, No. 042001.
(46) Soddemann, T.; Dünweg, B.; Kremer, K. Dissipative particle dynamics: A useful thermostat for equilibrium and nonequilibrium molecular dynamics simulations. *Phys. Rev. E* **2003**, *68*, No. 046702.
(47) Junghans, C.; Praprotnik, M.; Kremer, K. Transport properties controlled by a thermostat: An extended dissipative particle dynamics thermostat. *Soft Matter* **2008**, *4*, 156−161.
(48) Kobayashi, T.; Zhang, H.; Fukuzawa, K.; Itoh, S. Effect of transverse dissipative particle dynamics on dynamic properties of nanometer-thick liquid films on solid surfaces. *Mol. Simul.* **2020**, *46*, 1281−1290.

(49) Ren, L. Z.; Hu, H. B.; Bao, L. Y.; Xie, L.; Wen, J. Transverse effect on liquid viscosity: A many-body dissipative particle dynamics simulation study. *Phys. Fluids* **2022**, *34*, No. 012006.
(50) Gao, P.; Feng, J. J. Enhanced slip on a patterned substrate due to depinning of contact line. *Phys. Fluids* **2009**, *21*, No. 102102.

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